FRANCISZEK FRASZCZYK

MEng Mechatronics Engineering · Dublin City University

franciszekfraszczyk@gmail.com | +353 87 135 3161 | Kilcullen, Co. Kildare

EDUCATION

MEng Mechatronics
Dublin City University
2021 – 2026

BEng Mechatronics
Dublin City University
2021 – 2026

Leaving Certificate
Cross & Passion College
542 CAO Points · 2021

TECHNICAL SKILLS

PLC & Automation
- PLC Programming
- SCADA Screen Design
- HMI Networking
- Commissioning

AI & Embedded Systems
- TensorFlow Lite Micro
- Edge AI
- Convolutional Neural Networks (CNN)
- ESP32 Microcontroller
- Arduino Microcontroller
- MQTT Communication

Programming
- Python
- C / C++
- MATLAB
- OOP

Engineering
- Technical Drawings
- Circuit Analysis
- Software Development
- Data Analytics
- System Simulation & Control
- Sustainability

Word & Data Processing
- Microsoft Office Suite
- Google Workspace

PROFILE

Mechatronics engineer with a BEng and MEng from DCU, specialising in automation, embedded systems, and AI-driven industrial solutions. Hands-on experience commissioning live industrial plants, programming PLCs, and deploying machine learning models on resource-constrained hardware. Proven ability to bridge software and hardware challenges in real-world manufacturing environments.

EXPERIENCE

Automation Intern GEA Group *Mar 2024 – Dec 2024*

- Commissioned systems on a live industrial plant, gaining direct exposure to safety-critical operational environments.
- Programmed and ran PLC code to control industrial processes, from initial setup to full operational capacity.
- Designed and implemented SCADA screens for operator user interfaces, improving process visibility on the plant floor.
- Connected and configured HMIs to the plant network, ensuring reliable communication across control systems.
- Conducted pressure testing of pipework and upheld site health & safety protocols throughout the placement.
- Worked across multiple machine types, building broad practical knowledge of industrial automation systems.

MENG THESIS

Predictive Maintenance Using Physical AI DCU - Supervisor: Dr. Derek Molloy

Designed and built a complete end-to-end predictive maintenance (PdM) system for industrial machinery, combining embedded hardware with deep learning inference on the edge.

- Developed vibration-based anomaly detection using a triaxial accelerometer mounted on an experimental motor rig.
- Iteratively designed and trained a 1D Convolutional Neural Network (1D-CNN), progressing from a regression baseline to a quantised deep learning model.
- Created physically inspired anomaly augmentation techniques to generate bearing-fault signatures in the absence of labelled fault data.
- Deployed the quantised model on an ESP32-S3 microcontroller using TensorFlow Lite Micro, achieving real-time inference with minimal memory footprint.
- Implemented MQTT-based remote monitoring for lightweight transmission of anomaly alerts across the network.
- Demonstrated robust detection of mechanical resonance and electrically induced imbalance faults across multiple operating speeds.

LANGUAGES

- English – Native
- Polish – Native

CERTIFICATIONS

- SOLAS Safe Pass
- RSA EU Driver License
- CSWA
- Food & Beverage Safety

LinkedIn:

ADDITIONAL WORK EXPERIENCE

Retail Assistant Circle K *Mar 2025 – Sep 2025*

- Customer facing role in a high-volume forecourt environment.
- Operated tills whilst ensuring transactions, log balances and adequate float is maintained.
- Ensured optimal stock rotation to minimise waste.
- Responsible for maintenance of shop floor.

Bartender Killashee Hotel & Spa, FBD Hotels *2022 – 2024*

- Bartender across weddings, restaurant, bistro and function events.
- Trained new staff members and assisted with event setup, reservation management, and bar operations.
- Stock clerk responsibilities including ordering, stock rotation management, and restocking bars.
- Resolved escalated customer queries in a timely and respectful manner.

INETEREST & ACTIVITIES

- Trinity College Dublin Walton Club alumnus.
- Personal Arduino electronics projects.
- Programming.
- Fitness (Fencing, Hiking, Swimming).